Latent Representations for Better Generative Visual Modeling

Spyros Gidaris, Senior Research Scientist

50th Pattern Recognition & Computer Vision Colloquium
Czech Technical University, Prague
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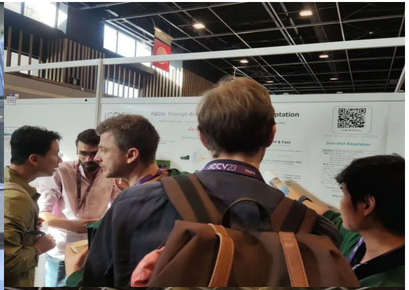
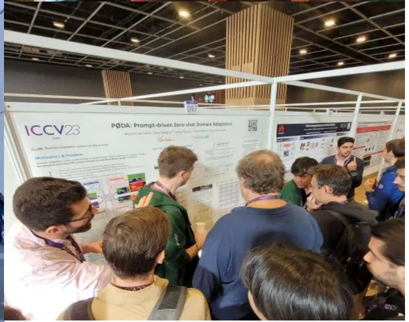
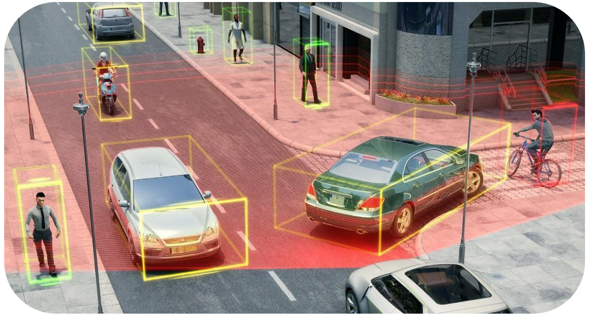


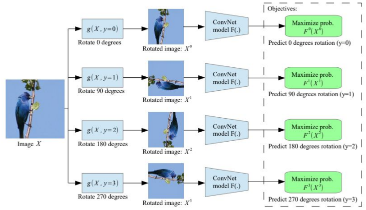




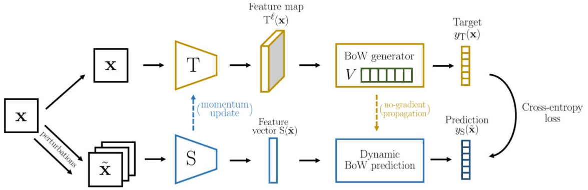
- ~25 researchers & PhDs
- Dedicated to open research
- 10s of academic collabs across France and Europe
- Offices: Paris, Prague
- Topics: perception, data efficiency, forecasting, reliability, explainability

https://valeoai.github.io

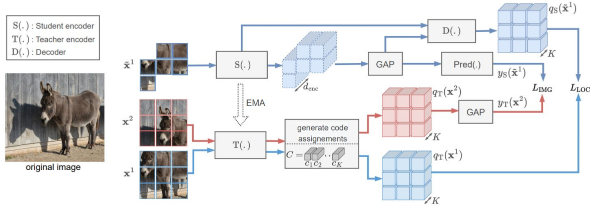
My main research interest the last few years: self-supervised visual representation learning



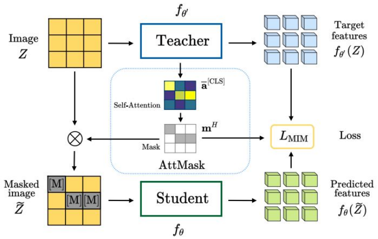
RotNet, Gidaris et al., ICLR'18



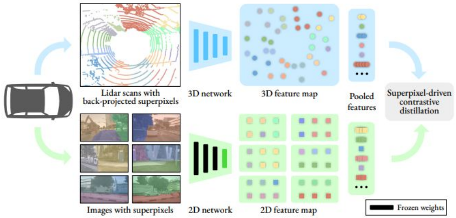
BoWNet, Gidaris et al., CVPR'20
OBoW, Gidaris et al. CVPR'21



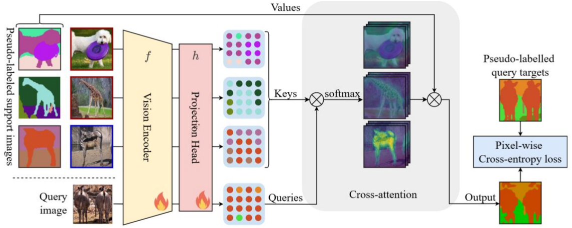
Gidaris et al., TMLR'24



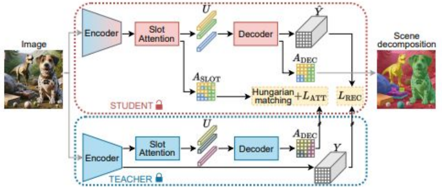
AttMask, Kakogeorgio et al., ECCV'22



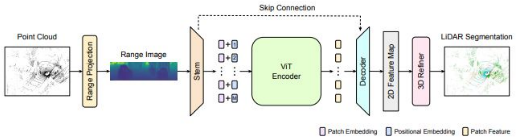
SLidR, Sautier et al., CVPR'22,
ScaLR, Puy et al. CVPR'24



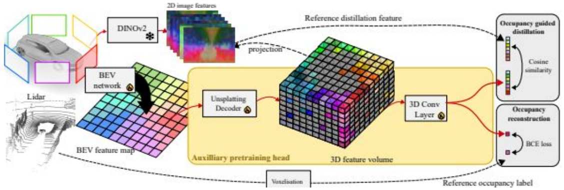
Sirko-Galouchenko et al., ICCV'25



SPOT, Kakogeorgio et al., CVPR'24



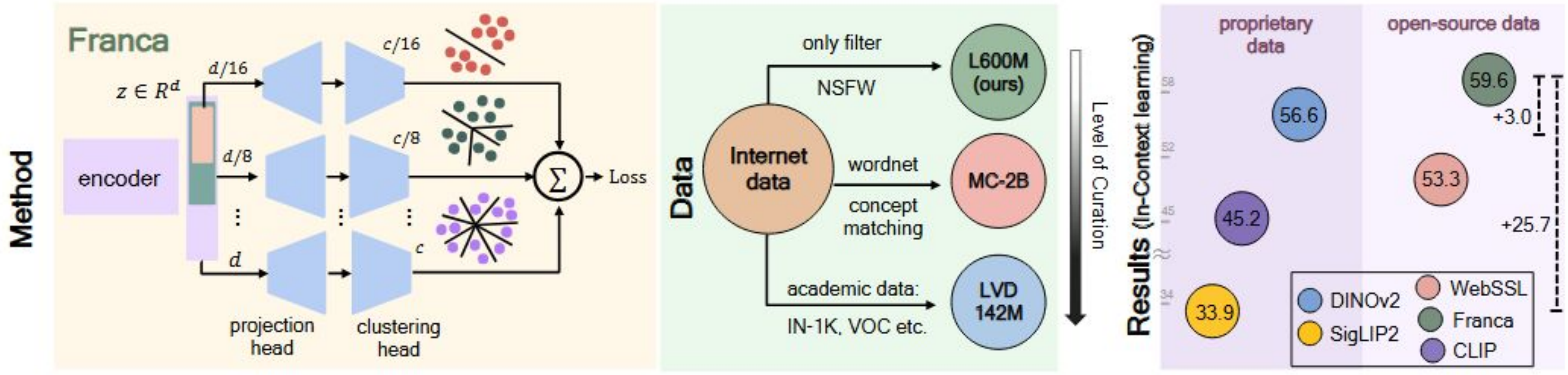
RangeViT, Ando et al., CVPR'23



Sirko-Galouchenko et al., CVPRW'24

`Franca`: a high-performing fully-open source (data, code, weights) self-supervised vision foundation model

https://github.com/valeoai/Franca



"Franca: Nested Matryoshka Clustering for Scalable Visual Representation Learning", Venkataramanan et al., 2025

Other recent work

DIP: Unsupervised Dense In-Context Post-training of Visual Representations

Sophia Sirko-Galouchenko[1,2] Spyros Gidaris[1]
Antonin Vobecky[1,3,4] Andrei Bursuc[1] Nicolas Thome[2]

[1]Valeo.ai [2]Sorbonne Université, CNRS, ISIR, F-75005 Paris, France
[3]FEE CTU [4]CIIRC CTU Prague

Figure 1. **Dense retrieval-based semantic segmentation in low-shot regimes (40 examples) using patch feature similarities.** For each query image (a), we show the ground truth (b), our DIP model's prediction (c), and its retrieved neighbor (d) for the patch marked with a red cross in (a). In (e) and (f), we display DINOv2R's (Dv2R) output and its nearest neighbor, respectively. Our DIP representations retrieve more coherent neighbors, yielding more accurate segmentations than DINOv2R.

ICCV 2025

Multi-Token Prediction Needs Registers

Anastasios Gerontopoulos[1,3] Spyros Gidaris[2] Nikos Komodakis[1,3,4]

[1]Archimedes, Athena Research Center
[2]valeo.ai
[3]University of Crete
[4]IACM-Forth

Abstract

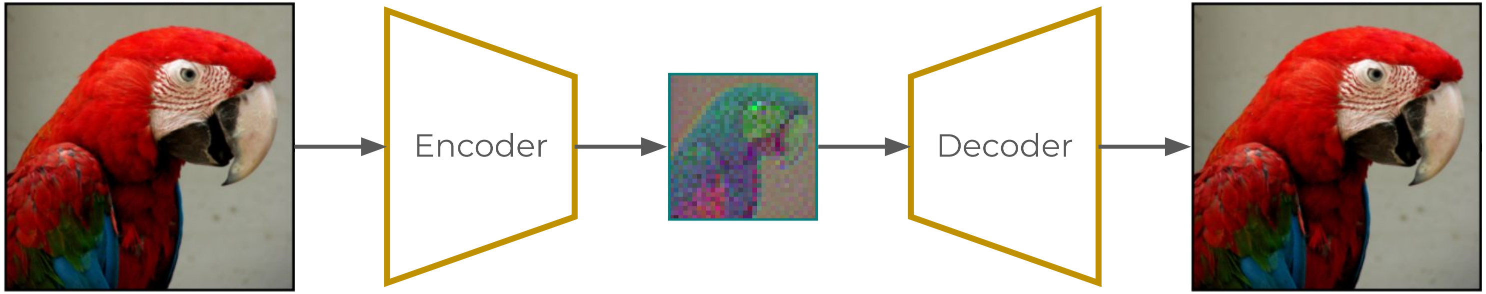
Multi-token prediction has emerged as a promising objective for improving language model pretraining, but its benefits have not consistently generalized to other settings such as fine-tuning. In this paper, we propose MuToR, a simple and effective approach to multi-token prediction that interleaves learnable register tokens into the input sequence, each tasked with predicting future targets. Compared to existing methods, MuToR offers several key advantages: it introduces only a negligible number of additional parameters, requires no architectural changes—ensuring compatibility with off-the-shelf pretrained language models—and remains aligned with the next-token pretraining objective, making it especially well-suited for supervised fine-tuning. Moreover, it naturally supports scalable prediction horizons. We demonstrate the effectiveness and versatility of MuToR across a range of use cases, including supervised fine-tuning, parameter-efficient fine-tuning (PEFT), and pretraining, on challenging generative tasks in both language and vision domains. Our code will be available at: https://github.com/nasosger/MuToR.

NeurIPs 2025

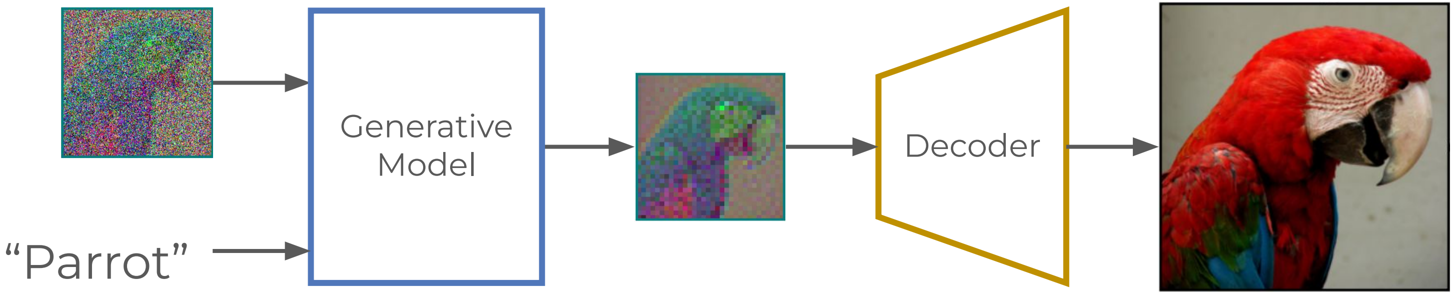
But this talk is about what (self-supervised) representation learning can do for generative visual modeling

Latent Generative Models: A Two-Stage Pipeline

Stage 1: VAE maps data to a compressed latent space

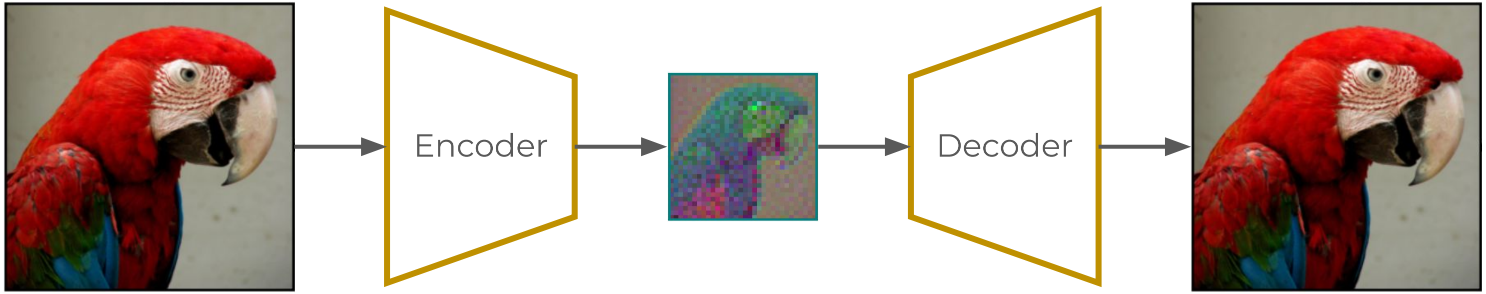


Stage 2: Generative model generates in the compressed latent space



“High-Resolution Image Synthesis with Latent Diffusion Models”, Rombach et al., 2022

Latent Representations



Critical role to the overall generative model's performance:

- **Compressed latent space**: lower resolution ⇒ more efficient
- **Remove high-frequency details:** easier to model & faster sampling

"High-Resolution Image Synthesis with Latent Diffusion Models", Rombach et al., 2022

What is a good latent space for generative visual modeling?

EQ-VAE: Equivariance Regularized Latent Space for Improved Generative Image Modeling

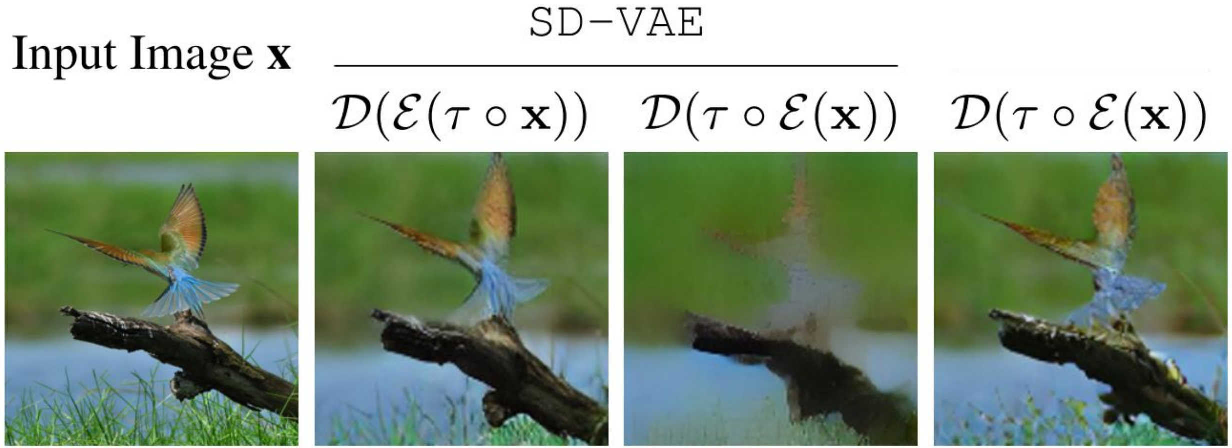
Theodoros Kouzelis, Ioannis Kakogeorgiou, Spyros Gidaris, Nikos Komodakis

ICML 2025

https://github.com/zelaki/eqvae

Motivation: VAE's Latent Space Lacks Equivariance

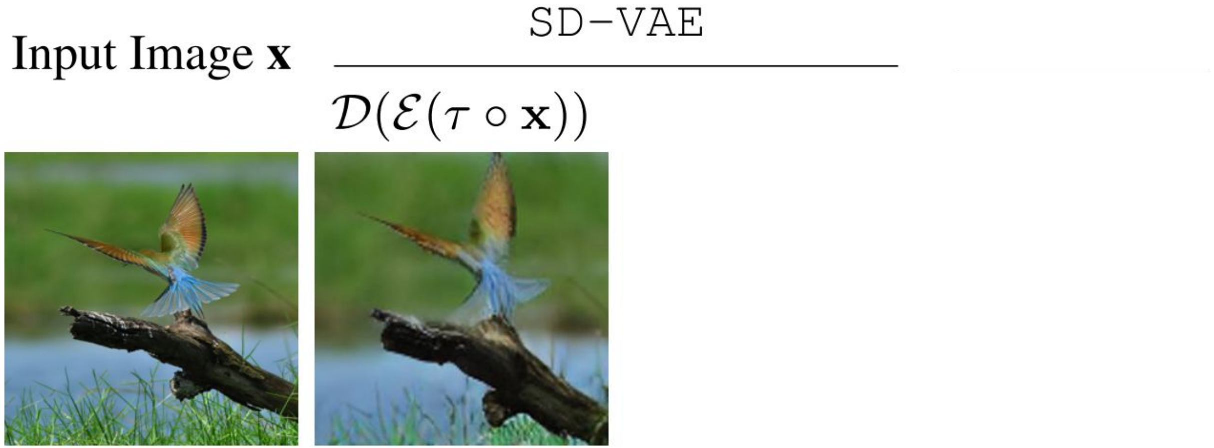
State-of-the-art VAEs produce latent representations that lack equivariance under basic spatial transformations like scaling



Motivation: VAE's Latent Space Lacks Equivariance

Input Image $\mathbf{x}$

Motivation: VAE's Latent Space Lacks Equivariance

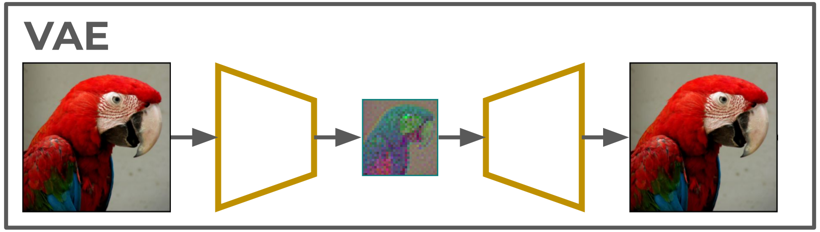


Motivation: VAE’s Latent Space Lacks Equivariance

VAE’s Decoder

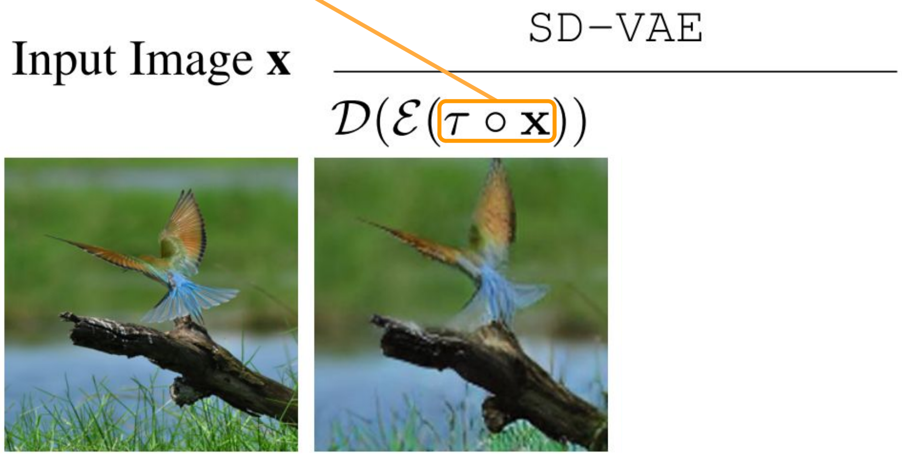
VAE’s Encoder

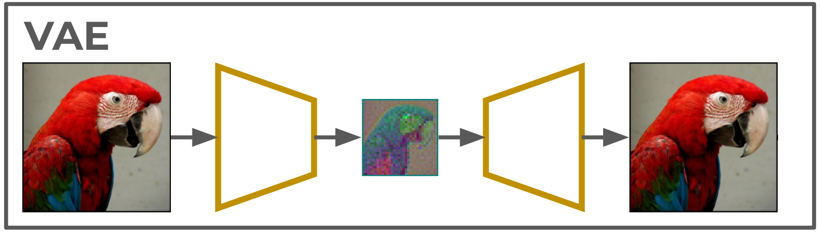
Input Image $\mathbf{x}$	SD-VAE
	$\boxed{\mathcal{D}}(\boxed{\mathcal{E}}(\tau \circ \mathbf{x}))$



Motivation: VAE's Latent Space Lacks Equivariance

Scaling the input

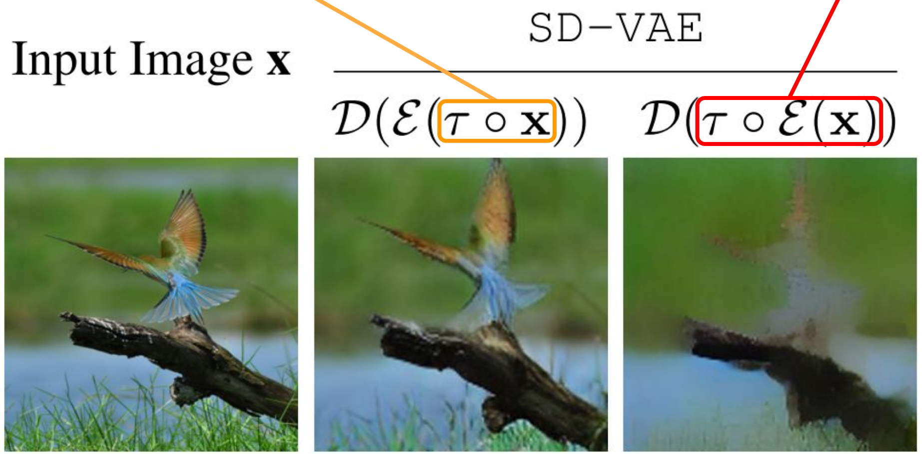


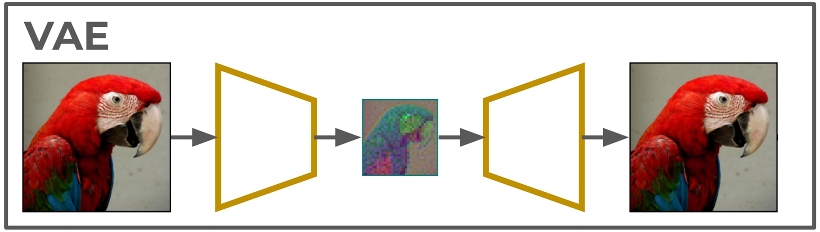


Motivation: VAE's Latent Space Lacks Equivariance

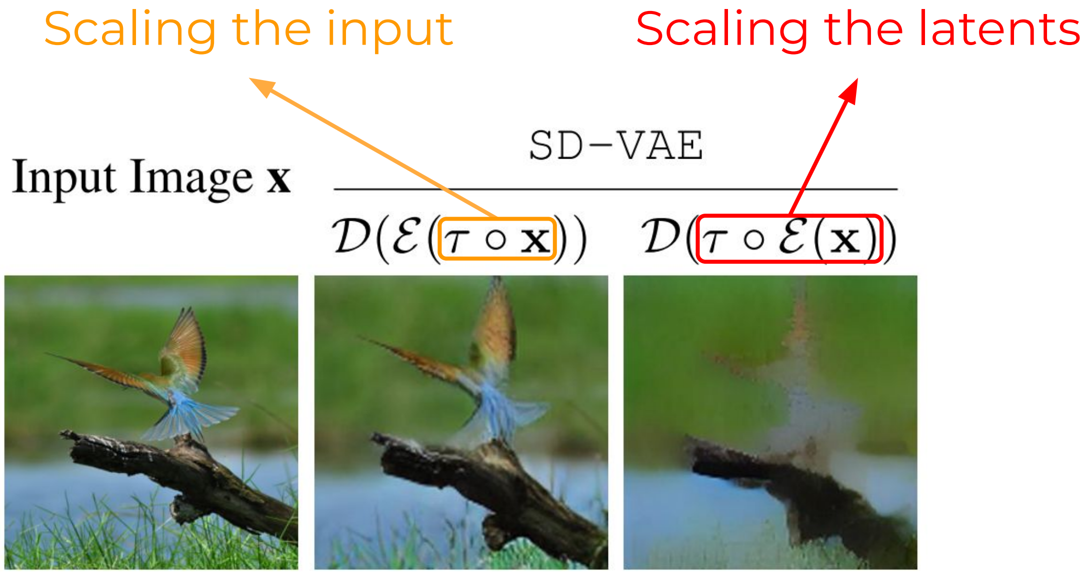
Scaling the input

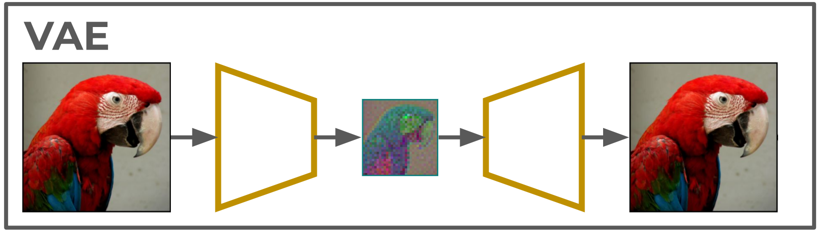
Scaling the latents





Motivation: VAE’s Latent Space Lacks Equivariance

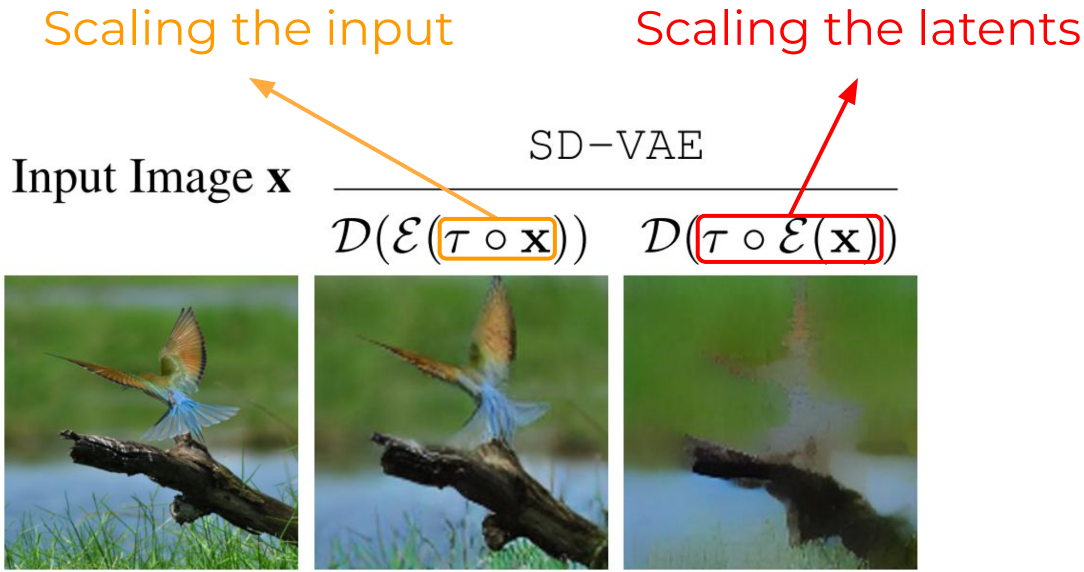




Equivariance A latent representation $\mathcal{E}(\mathbf{x})$ is equivariant with a transformation τ of the input image $\mathbf{x}$ if the transformation can be transferred to the representation output:

$$\forall \mathbf{x} \in \mathcal{X} : \quad \mathcal{E}(\tau \circ \mathbf{x}) = \tau \circ \mathcal{E}(\mathbf{x}). \tag{3}$$

Motivation: VAE’s Latent Space Lacks Equivariance

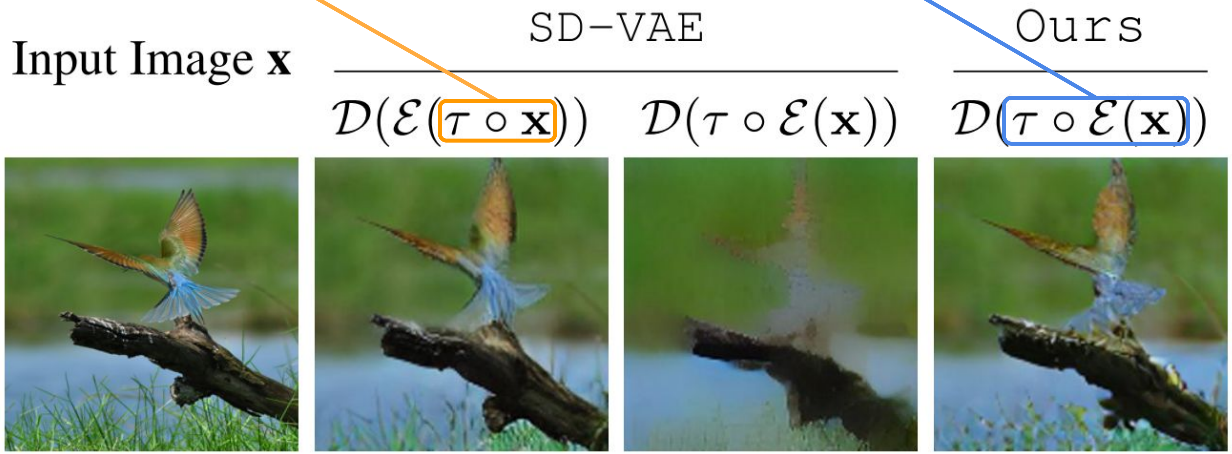


- Input transformations are not preserved in the latent space ⇒
- **Increases latent space complexity**

If we regularize the latent space to be equivariant, would that lead to better generative modeling?

Scaling the input

Scaling the latents

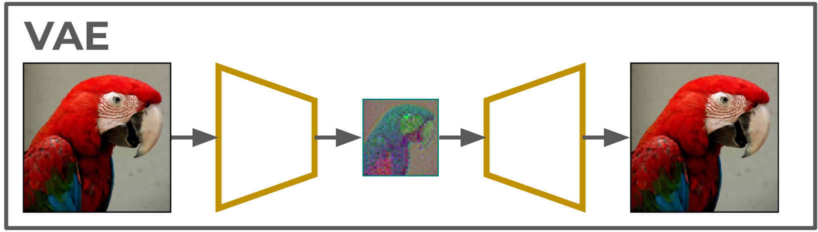


How do we enforce equivariance?

EQ-VAE: Regularization via equivariance constraints

Standard VAE training objective:

$$\mathcal{L}_{\mathrm{VAE}}(\mathbf{x}) = \mathcal{L}_{rec}\Big(\mathbf{x}, \mathcal{D}\big(\mathcal{E}(\mathbf{x})\big)\Big) + \lambda_{gan}\mathcal{L}_{gan}\Big(\mathcal{D}\big(\mathcal{E}(\mathbf{x})\big)\Big) + \lambda_{reg}\mathcal{L}_{reg}$$



EQ-VAE: Regularization via equivariance constraints

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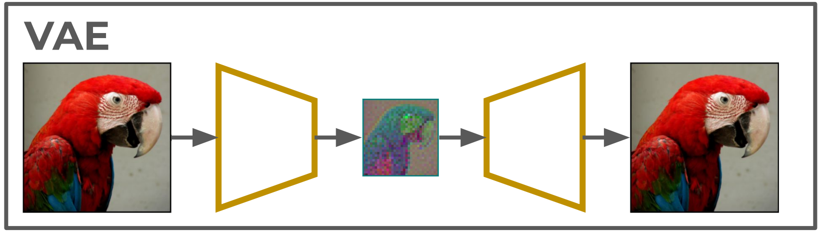
EQ-VAE training objective:

$$\mathcal{L}_{\text{EQ-VAE}}(\mathbf{x}, \tau) = \mathcal{L}_{rec}\Big(\boxed{\tau \circ}\,\mathbf{x}, \mathcal{D}\big(\boxed{\tau \circ}\,\mathcal{E}(\mathbf{x})\big)\Big) + \lambda_{gan}\mathcal{L}_{gan}\Big(\mathcal{D}\big(\boxed{\tau \circ}\,\mathcal{E}(\mathbf{x})\big)\Big) + \lambda_{reg}\mathcal{L}_{reg}$$

$$\mathcal{L}_{total}(\mathbf{x}) = \begin{cases} \mathcal{L}_{\text{VAE}}(\mathbf{x}) & p < p_\alpha, \\ \mathcal{L}_{\text{EQ-VAE}}(\mathbf{x}, \tau) & p \geq p_\alpha. \end{cases}$$



Spatial transformations used in EQ-VAE

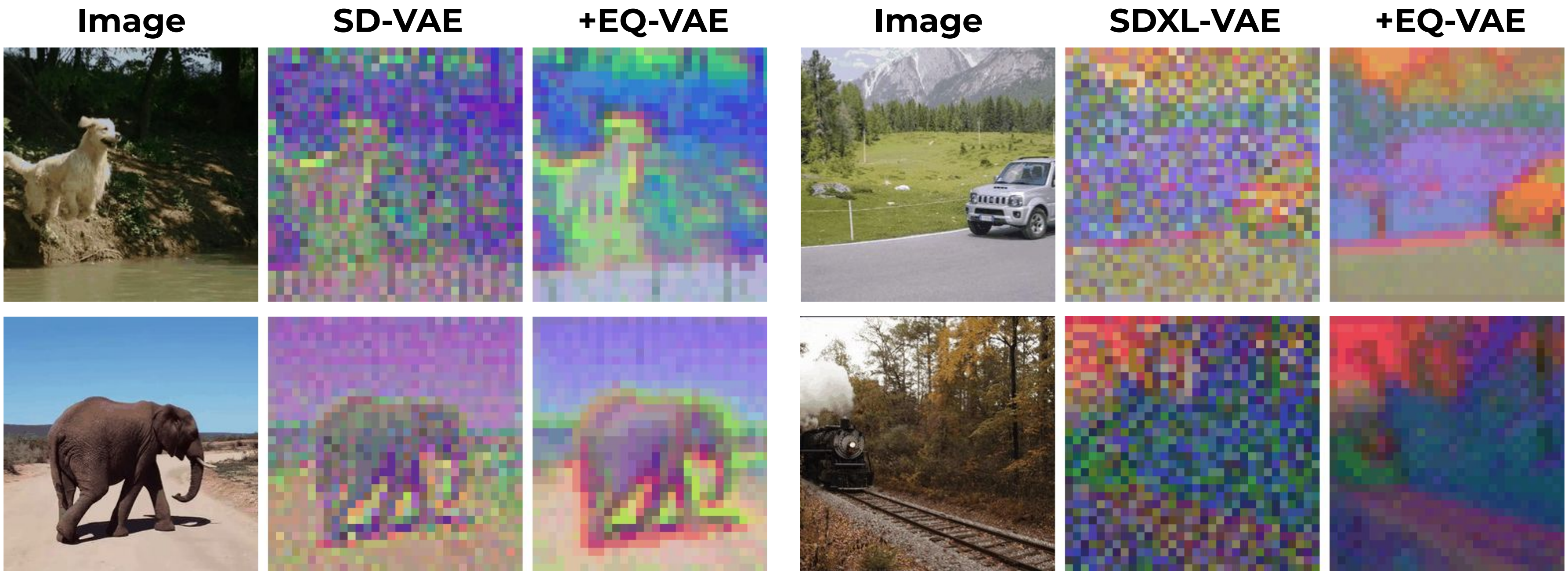
EQ-VAE training objective:

$$\mathcal{L}_{\text{EQ-VAE}}(\mathbf{x}, \tau) = \mathcal{L}_{rec}\Big(\boxed{\tau \circ} \mathbf{x}, \mathcal{D}\big(\boxed{\tau \circ} \mathcal{E}(\mathbf{x})\big)\Big) + \lambda_{gan}\mathcal{L}_{gan}\Big(\mathcal{D}\big(\boxed{\tau \circ} \mathcal{E}(\mathbf{x})\big)\Big) + \lambda_{reg}\mathcal{L}_{reg}$$

Two types of transformations:

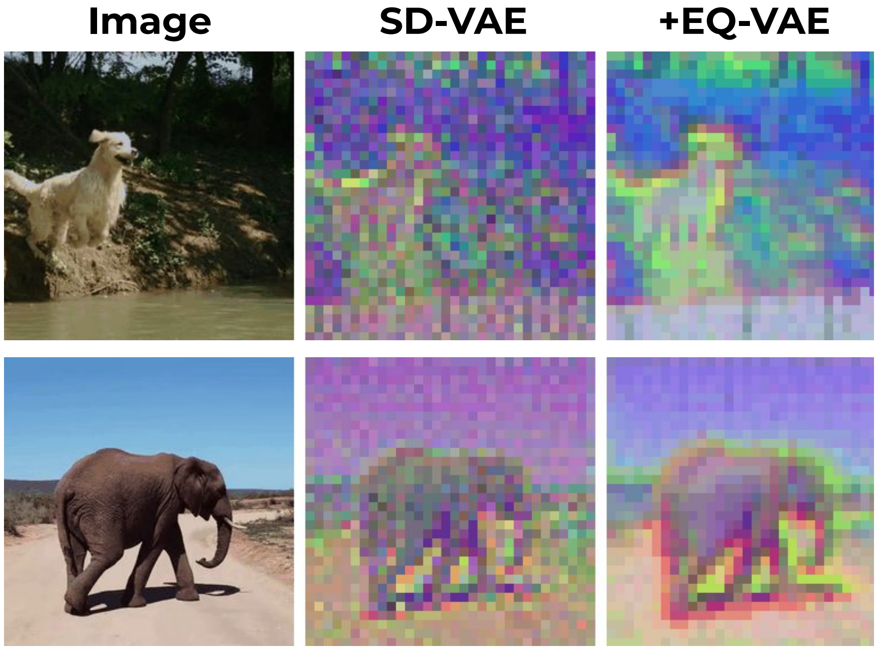
1. Isotropic scaling between 0.25 and 1.0
2. Rotations of multiples of 90 degrees

EQ-VAE produces “smoother” latent representations …



PCA visualizations of latent features

… without degrading the reconstruction quality

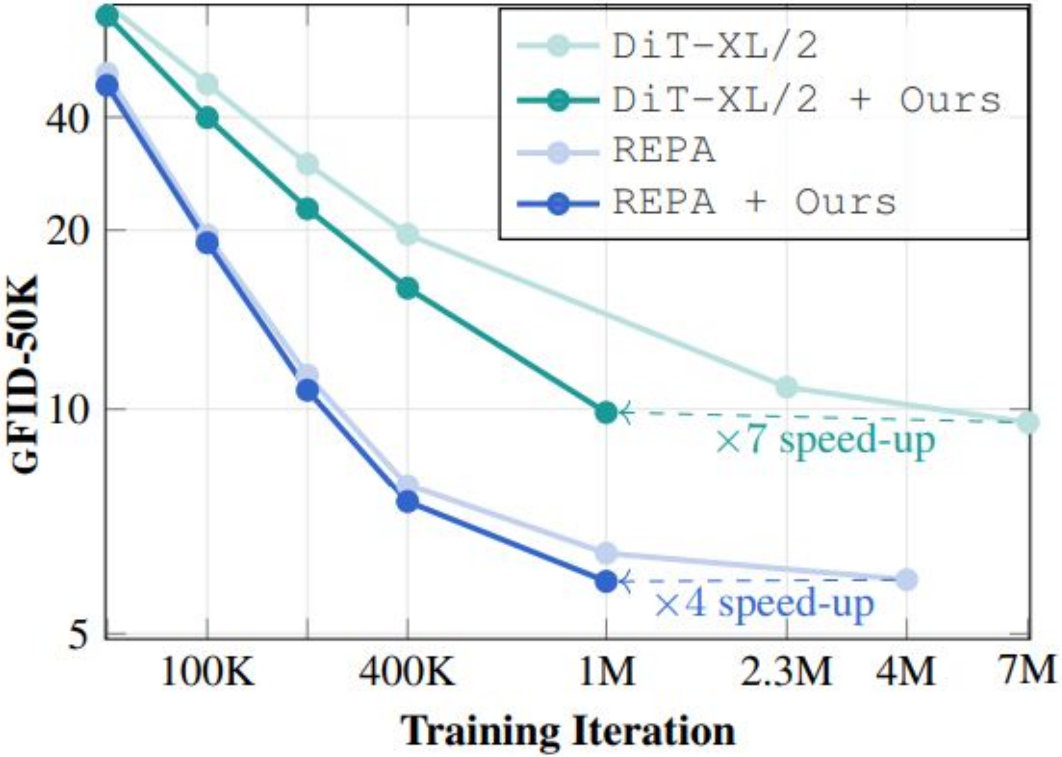


PCA visualizations of latent features

AUTOENCODER	RFID↓
SD-VAE	0.90
+ EQ-VAE (ours)	0.82
SDXL-VAE	0.67
+ EQ-VAE (ours)	0.65
SD3-VAE	0.20
+ EQ-VAE (ours)	0.19
SD-VAE-16	0.87
+ EQ-VAE (ours)	0.82

Reconstruction FID

EQ-VAE latents are easier for diffusion models to learn



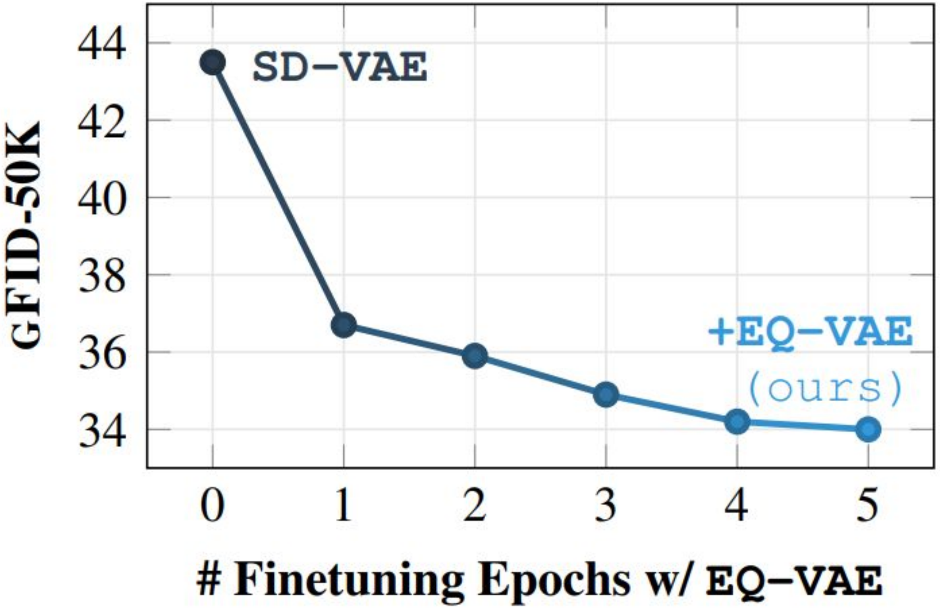
DiT: “Scalable Diffusion Models with Transformers”, Peebles et al., ICCV’23
REPA: “Representation Alignment for Generation: Training Diffusion Transformers is Easier Than You Think”, Sihyun Yu et al., ICLR’25

System-level comparisons on ImageNet with CFG

MODEL	EPOCHS	gFID↓	sFID↓	IS↑	PRE.↑	REC.↑
LDM	200	3.60	-	247.7	0.87	0.48
MaskDiT	1600	2.28	5.67	276.6	0.80	0.61
SD-DiT	480	3.23	-	-	-	-
SiT-XL/2	1400	2.06	4.50	270.3	0.82	0.59
DiT-XL/2	1400	2.27	4.60	278.2	0.83	0.57
DiT-XL/2 †	1400	2.47	5.18	276.1	0.82	0.57
+ EQ-VAE (ours)	300	2.37	4.78	277.3	0.82	0.57
REPA*	800	1.42	4.70	305.7	0.80	0.65
+ EQ-VAE * (ours)	200	1.70	5.13	283.0	0.79	0.62

Table 4: **Comparison on ImageNet 256×256 with CFG**. † indicates that the used autoencoder is the original SD-VAE (instead of SD-VAE-FT-EMA). REPA uses SiT-XL/2. * denotes that guidance interval (Kynkäänniemi et al., 2024) is applied.

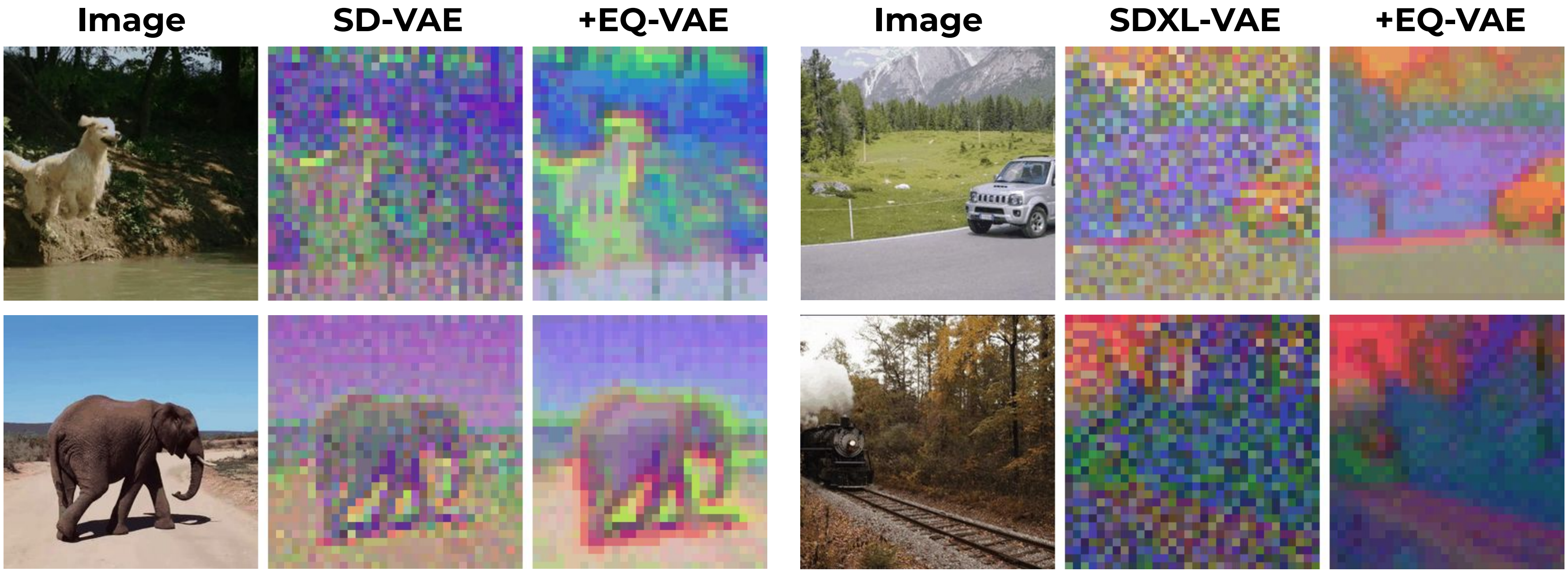
EQ-VAE requires just a very few fine-tuning epochs



Experimental setup:
Train a DiT-B/2 diffusion model from scratch after each fine-tuning epoch of EQ-VAE

DiT: "Scalable Diffusion Models with Transformers", Peebles et al., ICCV'23

EQ-VAE — Sum up



- Established VAEs for latent generative models **lack equivariance**
- EQ-VAE produces **easier-to-learn latent representations ⇒**
- **Improved convergence speed and generative performance**

Boosting Generative Image Modeling via Joint Image-Feature Synthesis

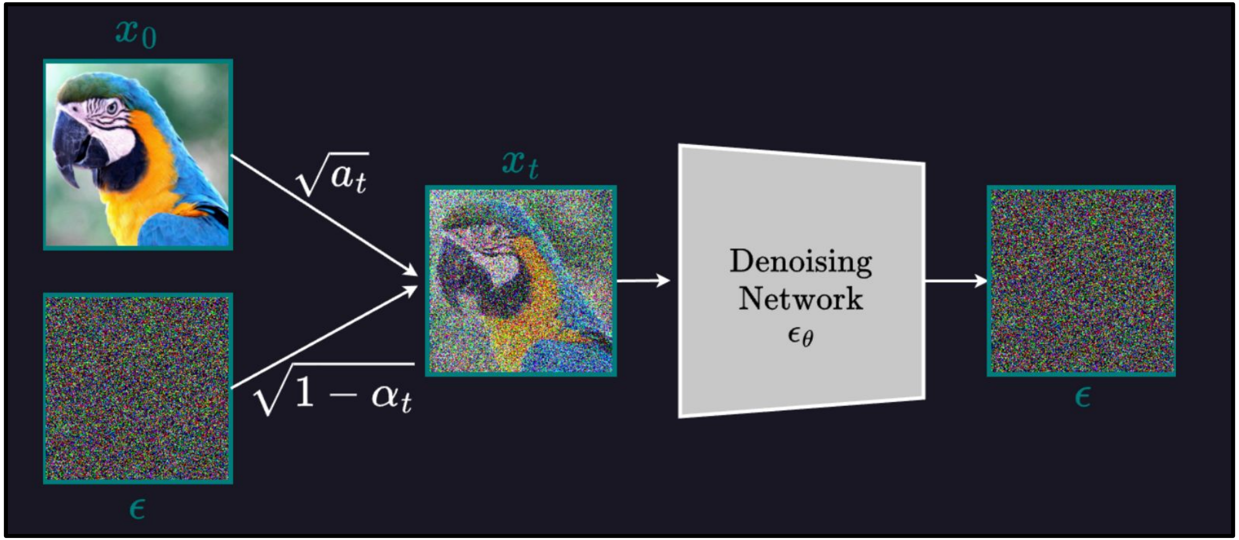
Theodoros Kouzelis, Eystathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, Nikos Komodakis

NeurIPs 2025

https://github.com/zelaki/ReDi

Beyond VAE latent design:
How else can we boost generation?
→ Enforcing semantic representations?

The problem with the Denoising Objective



- The goal is purely to remove noise, regardless of the content
- Doesn't distinguish between meaningful semantics and low-level details
- Does not result in (very) good representations like SSL methods

REPA: Distills pretrained SSL features into DiT features

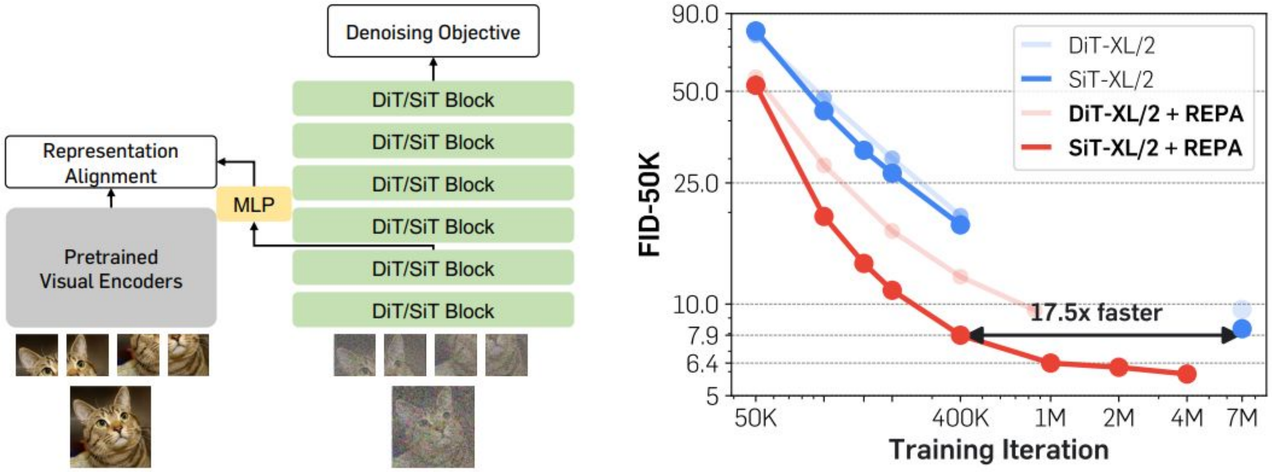


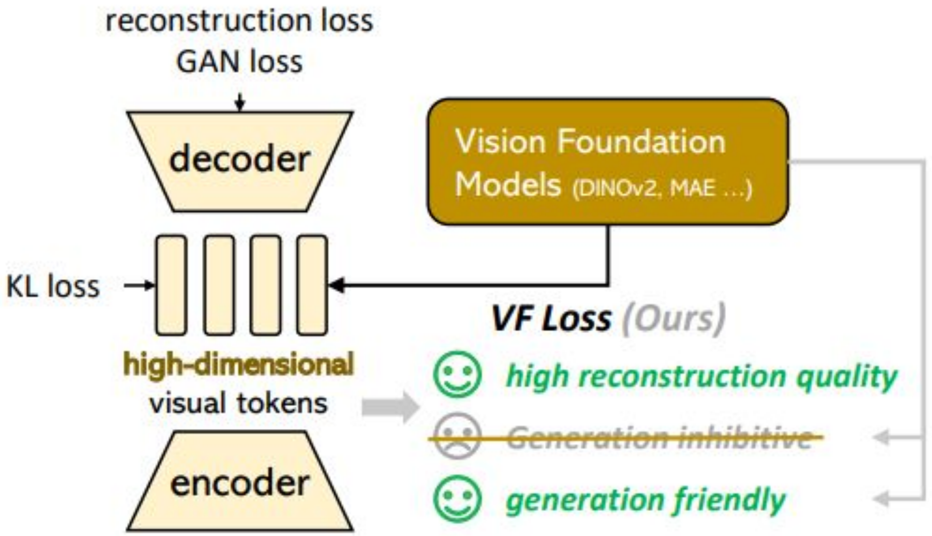
Figure 1: **Representation alignment makes diffusion transformer training significantly easier.**

Align intermediate DiT/SiT features with representations of powerful visual encoders (e.g., DINOv2)

“Representation Alignment for Generation: Training Diffusion Transformers is Easier Than You Think”, Sihyun Yu et al., ICLR’25

VA-VAE: representation alignment for VAE latents

Aligning the VAE latents with DINOv2 also results in better generative performance

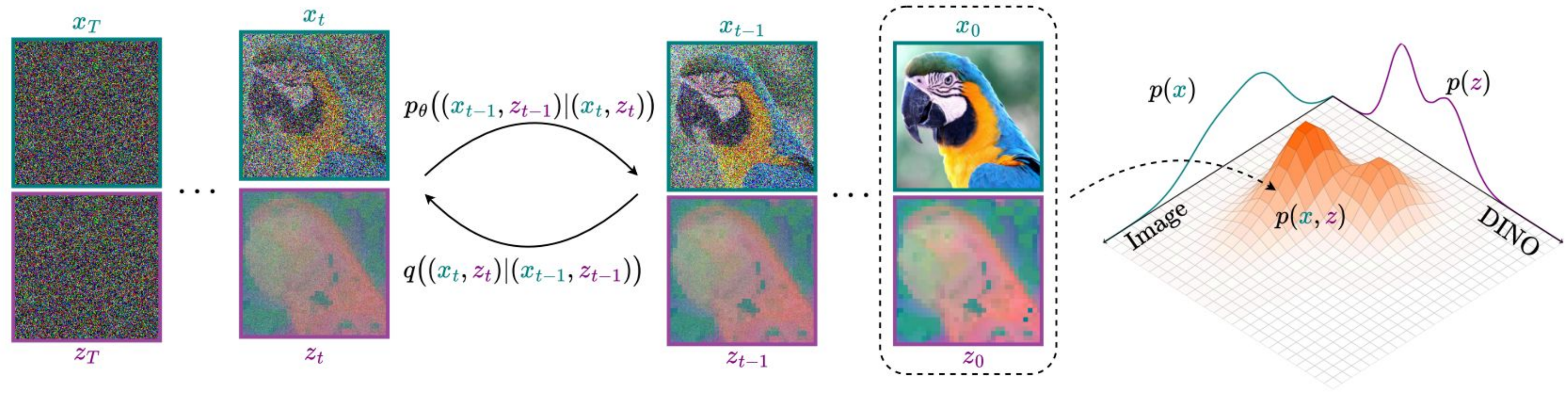


“Reconstruction vs. Generation: Taming Optimization Dilemma in Latent Diffusion Models ”, Yao et al., CVPR’25

Previous results establish a clear connection between representation learning and generative performance

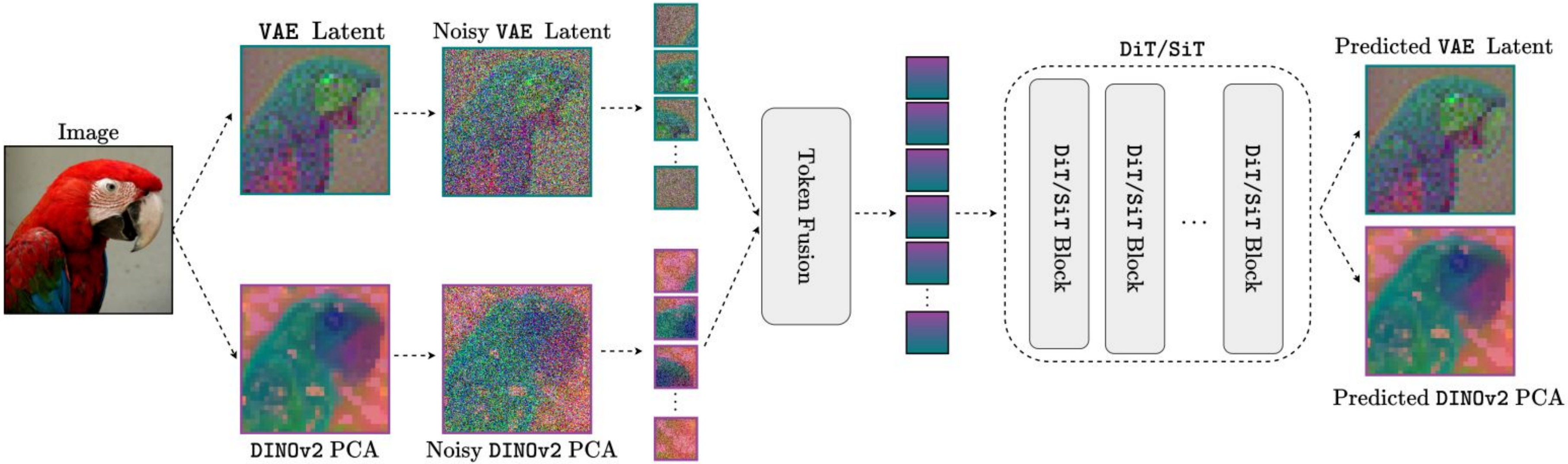
Is there a more effective approach for leveraging pretrained visual representations?

ReDi: Joint Image-Feature Synthesis



Explicitly learns the joint distribution of both precise low-level (VAE) and high-level semantic (DINOv2) features

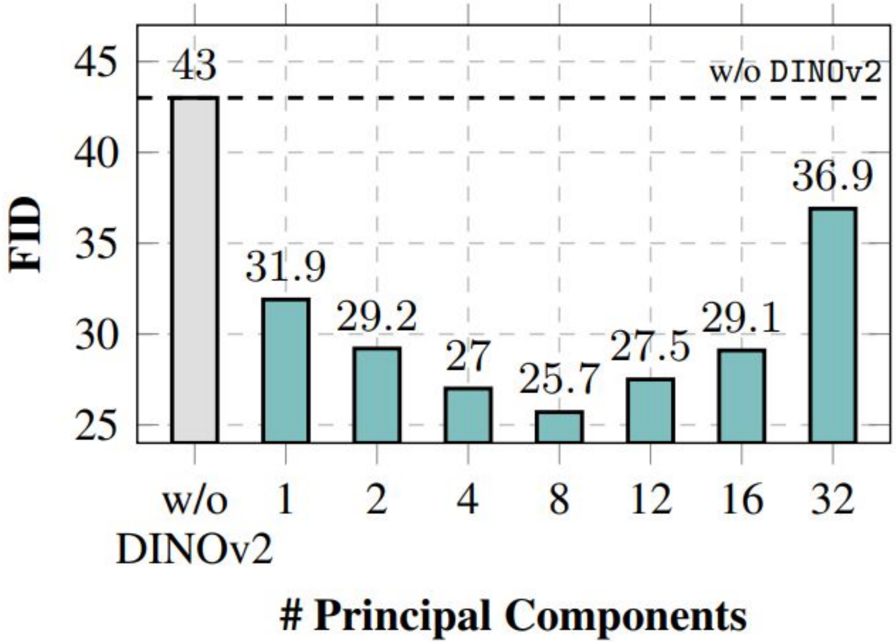
ReDi: Joint denoising both VAE latents and DINO feats



Dimensionality-Reduced Visual Representation

- DINOv2 channels (768) significantly exceed the VAE channels (4) ⇒ **DINOv2 dominates the capacity of the diffusion model.**

- PCA to reduce the DINOv2 dimensions to only 8 channels



Comparison with REPA
conditional results w/o CFG

Model	#Params	Iter.	FID↓
SiT-XL/2	675M	7M	8.3
w/ REPA	675M	4M	5.9
w/ ReDi (ours)	675M	700K	5.6
w/ ReDi (ours)	675M	4M	3.3

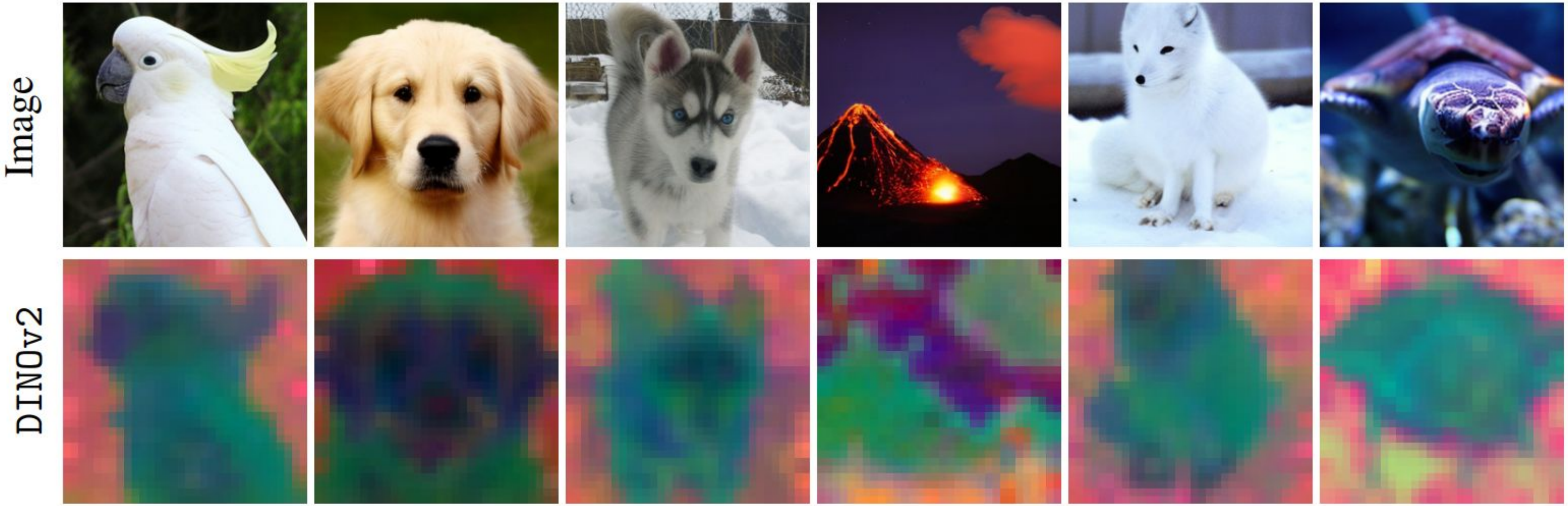
- Around x6 faster training convergence than REPA
- Better converged performance

System-level comparisons
conditional results w/ CFG

Table 2: **Comparison with State-of-the-art.** Quantitative evaluation on ImageNet 256×256 with Classifier-Free Guidance. Both REPA and ReDi (ours) employ SiT-XL/2 as the base model.

MODEL	EPOCHS	FID↓	sFID↓	IS↑	PRE.↑	REC.↑
			Autoregressive Models			
VAR	350	1.80	-	365.4	0.83	0.57
MagViTv2	1080	1.78	-	319.4	0.83	0.57
MAR	800	1.55	-	303.7	0.81	0.62
			Latent Diffusion Models			
LDM	200	3.60	-	247.7	0.87	0.48
U-ViT-H/2	240	2.29	5.68	263.9	0.82	0.57
DiT-XL/2	1400	2.27	4.60	278.2	0.83	0.57
MaskDiT	1600	2.28	5.67	276.6	0.80	0.61
SD-DiT	480	3.23	-	-	-	-
SiT-XL/2	1400	2.06	4.50	270.3	0.82	0.59
FasterDiT	400	2.03	4.63	264.0	0.81	0.60
MDT	1300	1.79	4.57	283.0	0.81	0.61
			Leveraging Visual Representations			
REPA	800	1.80	4.50	284.0	0.81	0.61
ReDi (ours)	350	1.72	4.68	278.7	0.77	0.63
ReDi (ours)	800	1.61	4.66	295.1	0.78	0.64

ReDi — Sum up



- Jointly models high-level semantic features and low-level VAE latents
- More effective ⇒ Better generative performance & convergence

ReDi is complementary to REPA

Table 5: **ReDi with REPA**. FID scores on ImageNet 256×256 w/o CFG.

MODEL	#ITER.	FID↓
SiT-XL/2 w/ REPA	4M	5.9
SiT-XL/2 w/ REPA+ReDi	350K	5.9
SiT-XL/2 w/ REPA+ReDi	1M	3.5

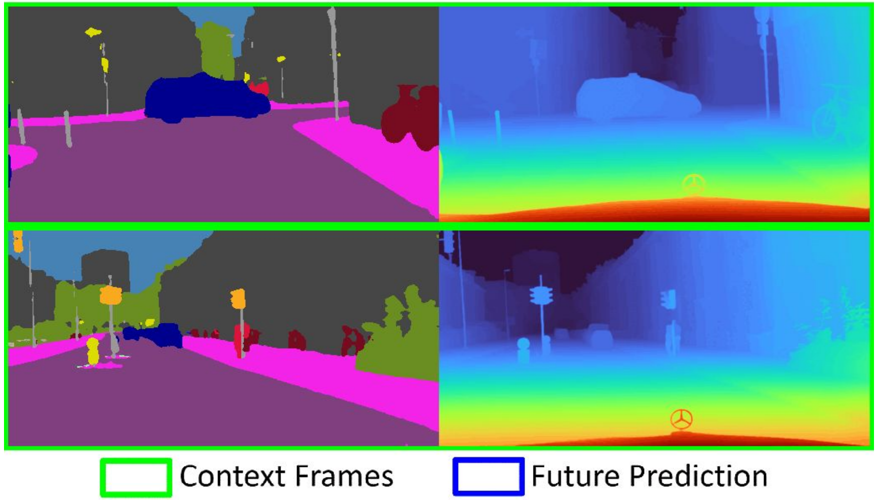
From image generation to future prediction

Credit: “Vista: A Generalizable Driving World Model with High Fidelity and Versatile Controllability”, Gao et al., NeurIPS’24

What representation space to use for future prediction?

Predicting the Future: RGB vs. Semantics

vs.



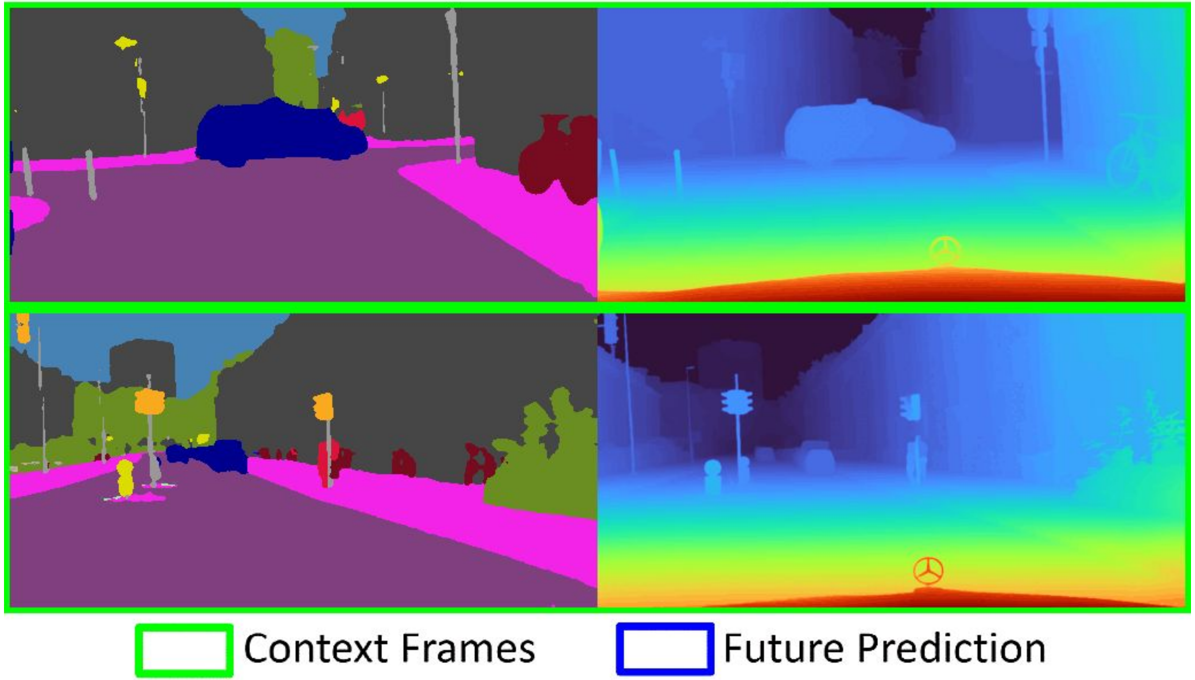
Why waste model capacity on pixels if what we want is semantic future states (what objects exist and where)?

CVPR 2025

Advancing Semantic Future Prediction through Multimodal Visual Sequence Transformers

Efstathios Karypidis[1,3] Ioannis Kakogeorgiou[1] Spyros Gidaris[2] Nikos Komodakis[1,4,5]

[1]Archimedes/Athena RC [2]valeo.ai
[3]National Technical University of Athens [4]University of Crete [5]IACM-Forth



DINO-Foresight: Looking into the Future with DINO

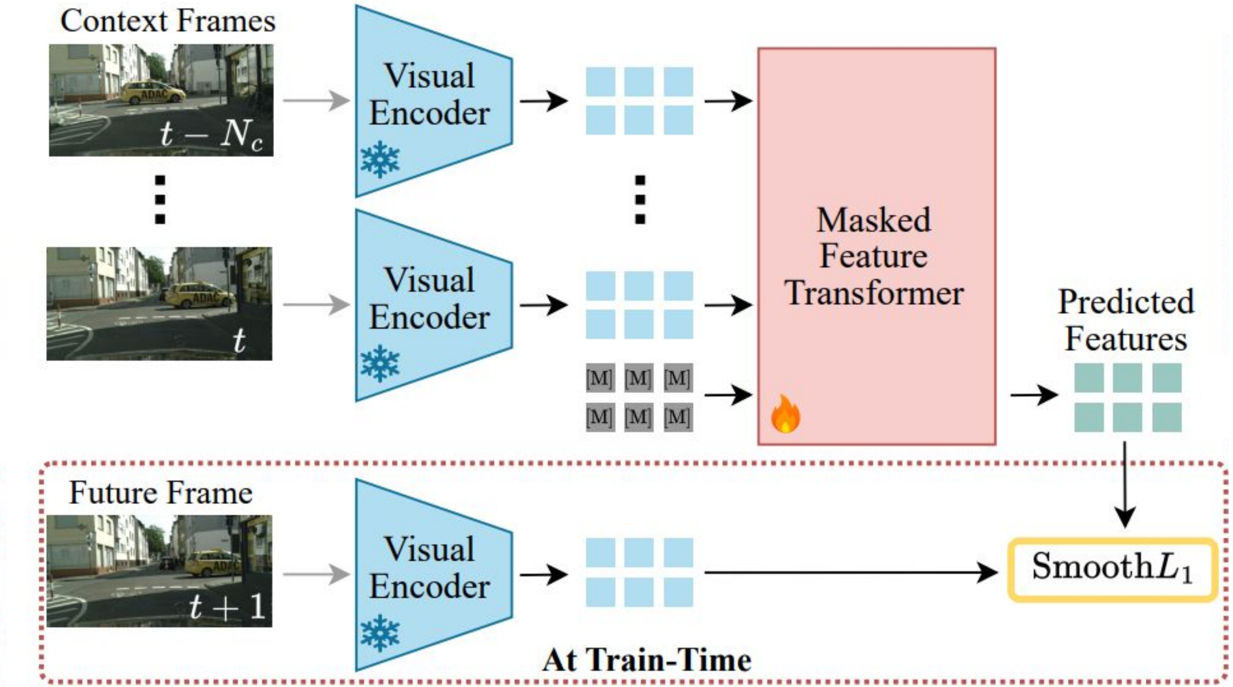
Eystathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, Nikos Komodakis

NeurIPs 2025

https://github.com/Sta8is/DINO-Foresight

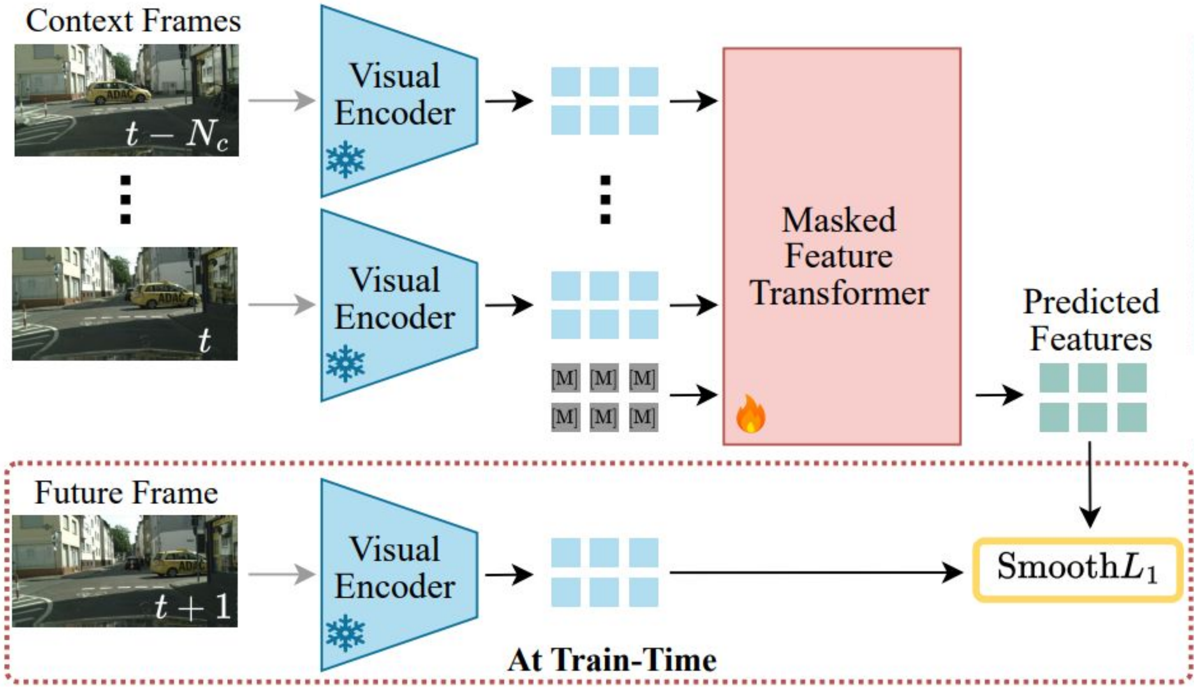
DINO-Foresight: Forecast SSL features

TL/DR: in video prediction replace the VAE latents with DINOv2 features



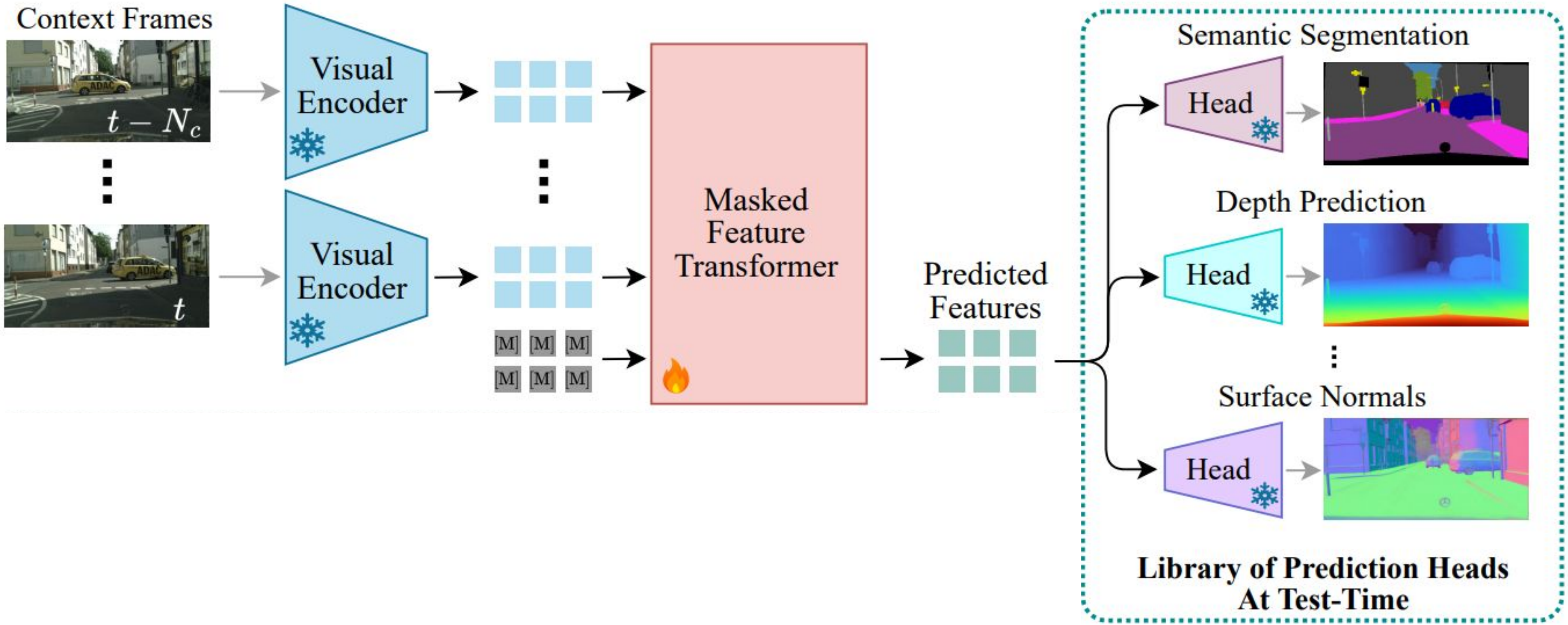
DINO-Foresight — Training

TL/DR: in video prediction replace the VAE latents with DINOv2 features

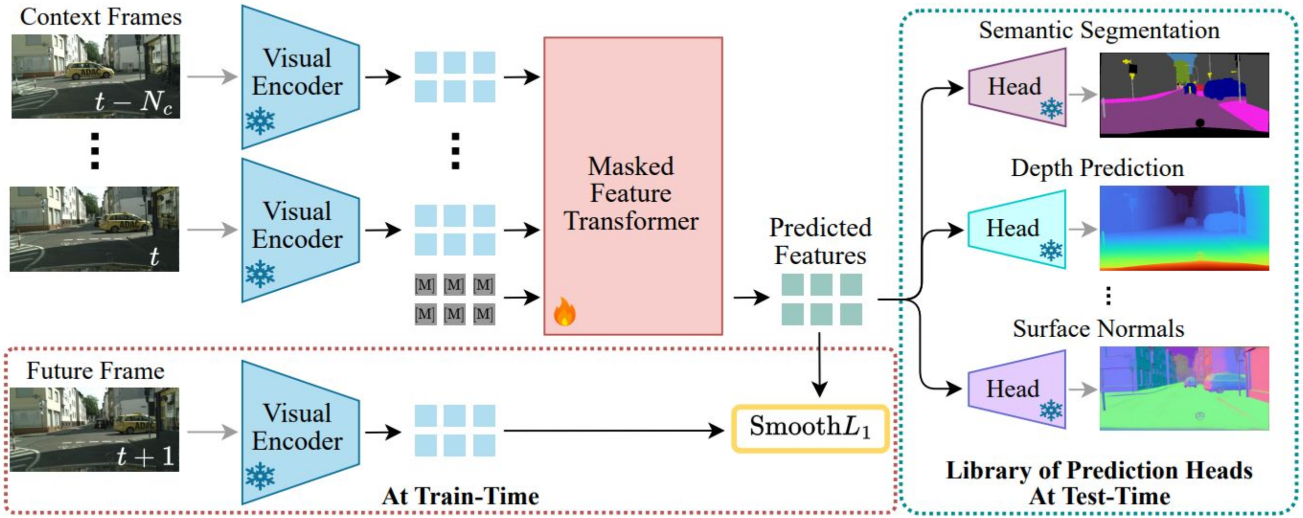


DINO-Foresight — Inference

Plug-and-play any prediction head for scene understanding tasks



DINO-Foresight: Forecast SSL features



1. **Leverages semantic representations**: strong scene understanding
2. **Avoids full-frame synthesis:** focus more on high-level changes
3. **Scalable multi-task support**: plug-and-play any prediction head

Future-frame scene understanding results

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

Cityscapes dataset

Future-frame scene understanding results

Experimental setup

Two prediction horizons

- **Short-term (0.18 sec)**
- Mid-term (0.54 sec)

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

Cityscapes dataset

Future-frame scene understanding results

Experimental setup

Two prediction horizons

- Short-term (0.18 sec)
- **Mid-term (0.54 sec)**

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

Cityscapes dataset

Future-frame scene understanding results

Experimental setup

Four scene understanding tasks for the future frames

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

Cityscapes dataset

Future-frame scene understanding results

Oracle baseline: use ground truth future features and applies prediction heads

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

Cityscapes dataset

Future-frame scene understanding results

DINO-Foresight: predicts future features and applies the appropriate task head

METHOD	SEMANTIC SEGMENTATION				INSTANCE SEGMENTATION				DEPTH		SURFACE NORMALS	
	SHORT		MID		SHORT		MID		SHORT	MID	SHORT	MID
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

Cityscapes dataset

Future-frame scene understanding results

VISTA baseline: video generation foundation model. Predict future frames with VISTA and apply perception models

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA $_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

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Future-frame scene understanding results

DINO-Foresight vs. **VISTA**

DINO-Foresight achieves **better results** with much **smaller inference time** (8mins for 500 scenes vs. several hours for VISTA).

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	29.2	12.9	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	60.3	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	48.7	24.9	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	71.8	71.7	59.8	57.6	**50.5**	**26.6**	27.3	12.6	88.6	85.4	**94.4**	**91.3**

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Future-frame scene understanding results

DINO-Foresight vs. **Prior <u>specialist</u> semantic future prediction works** (predict future features from one or multiple task-specific perception models)

DINO-Foresight is **on par with the best specialist** model while being able to **handles multiple tasks** at once

Method	Semantic Segmentation				Instance Segmentation				Depth		Surface Normals	
	Short		Mid		Short		Mid		Short	Mid	Short	Mid
	ALL	MO	ALL	MO	AP50	AP	AP50	AP	δ_1	δ_1	11.25°	11.25°
3Dconv-F2F-RGB	57.0	-	40.8	-	-	-	-	-	-	-	-	-
Dil10-S2S	59.4	55.3	47.8	40.8	-	-	-	-	-	-	-	-
F2F	-	61.2	-	41.2	39.9	19.4	19.4	7.7	-	-	-	-
ConvLSTM	60.1	-	-	-	-	-	-	-	-	-	-	-
FeatReproj3D	61.5	-	45.4	-	-	-	-	-	-	-	-	-
Bayesian S2S	65.1	-	51.2	-	-	-	-	-	-	-	-	-
3Dconv-F2F-SEG	65.5	-	50.5	-	-	-	-	-	-	-	-	-
DeformF2F	65.5	63.8	53.6	49.9	-	-	-	-	-	-	-	-
LSTM AM S2S	65.8	-	51.3	-	-	-	-	-	-	-	-	-
CPConvLSTM	-	-	-	-	44.3	22.1	25.6	11.2	-	-	-	-
APANet	-	64.9	-	51.4	46.1	23.2	<u>29.2</u>	<u>12.9</u>	-	-	-	-
LSTM M2M	67.1	65.1	51.5	46.3	-	-	-	-	-	-	-	-
IndRNN-Stack	67.6	60.8	58.1	52.1	-	-	-	-	-	-	-	-
DiffAttn-Fuse	67.9	61.2	58.1	51.7	-	-	-	-	-	-	-	-
F2MF	69.6	67.7	57.9	54.6	-	-	-	-	-	-	-	-
PFA (semantic)	71.1	69.2	<u>60.3</u>	56.7	-	-	-	-	-	-	-	-
PFA (instance)	-	-	-	-	<u>48.7</u>	<u>24.9</u>	**30.5**	**14.8**	-	-	-	-
Futurist	**73.9**	**74.9**	**62.7**	**61.2**	-	-	-	-	**96.0**	**91.9**	-	-
Oracle	77.0	77.4	77.0	77.4	66.2	40.4	66.2	40.4	89.1	89.1	95.3	95.3
VISTA$_{ft}$	64.9	62.1	53.9	51.0	33.1	17.7	19.8	9.0	86.4	82.8	93.0	90.0
DINO-Foresight (ours)	<u>71.8</u>	<u>71.7</u>	59.8	<u>57.6</u>	**50.5**	**26.6**	27.3	12.6	<u>88.6</u>	<u>85.4</u>	**94.4**	**91.3**

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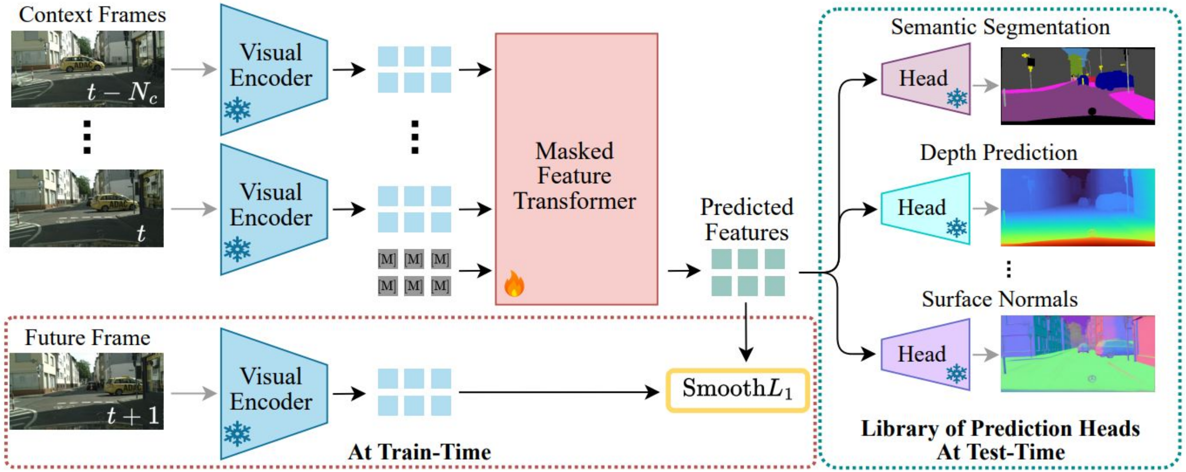
DINO-Foresight — sum-up



- Video prediction, but replace VAE latents with DINOv2 features
- Precise future scene understanding w/o modeling low-level details
- Unified model for diverse scene understanding tasks

To conclude

Better latent space design simplifies, accelerates, and improves generative visual modeling

- What is a good latent space for generative visual modeling?
- What representation learning can do for generative modeling?
- How to bridge the gap between representation learning and generative modeling?

Thank You!

https://gidariss.github.io/

valeo.ai